

Juwon Lee (Henna)

AI Product Engineer

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SUMMARY

An AI product engineer at the Seoul Robot & AI Science Museum who builds AI into working products, from the data pipeline to the deployed interface, and keeps them running where people use them: exhibition floors, classrooms, and hardware in the field. My Art Lab ran on the floor through June–July 2026, growing from 3,049 images to over 8,000; for Doctor-Green I built the training set myself, extracting 39,889 lesion crops with a group-aware stratified split.

SKILLS

AI & Computer Vision Python · TensorFlow.js · MobileNet v2 · VGG16 / CNN · OpenAI API

Data & Datasets VQA Datasets · Dataset QA · AI Hub · Stratified Splits

Frontend & App Next.js · React · TypeScript · JavaScript · Tailwind · PWA · Capacitor

Backend, Infra & Ops Supabase · Firebase / Firestore · Flask · Serverless Functions · Cloudflare R2 · Vercel · Tailscale / WireGuard

Hardware · IoT · Robotics ESP32 · MicroPython · Arduino / C++ · Sensors

Planning & Education Product Planning (PM) · Learning Design · Bilingual Docent (KO-EN)

EXPERIENCE

Education R&D · Developer

2026.03 – Present

Seoul Robot & AI Science Museum · Education Team · Seoul, KR

- Research, build, and run AI education web apps live on the exhibition floor; the Art Lab ran there for two months, growing from **3,049** images in June 2026 to over **8,000** in July.
- Design education programs end to end: a smart-farm curriculum commissioned by Yangpyeong Education Office, built with its full server and web-app stack, demo-taught to 5 high-school students in August 2026, with the **15**-student cohort class following in October 2026.
- Ship the internal ops tooling behind them, such as the staff scheduler used daily by the docent team.
- Run the Picabot robot-arm drawing studio as a regular 30-minute hands-on AI session for school groups, and led its extended 50-minute class for the 100-student KT × Seoul City AI Future Camp in August 2026.
- Guide exhibitions bilingually (Korean / English) and support the multipurpose education rooms.

VQA Dataset Builder & QA Reviewer

2025.11 – 2026.01

Select Star · Seoul, KR

- Built and reviewed VQA (visual question-answering) datasets for LLM training on an ETRI (Electronics and Telecommunications Research Institute) project, including complex 3-hop reasoning Q&A sets.
- Checked each question–response pair against the project's correction guidelines for grammatical and semantic accuracy, and rewrote assistant answers that did not follow logically from the user's question.
- Sustained **2.5×** the team's average throughput while holding accuracy.

Youth Intern

2025.04 – 2025.08

Ministry of Science and ICT · Sejong, KR

- Redesigned the national Pay-TV survey: defining what to measure and how to sample it, the same problem as designing a dataset.

- Covered 11 ministry exhibitions and conferences; authored reports and issue briefs.

SELECTED PROJECTS

Raimi's AI Art Lab *Seoul Robot & AI Science Museum · Exhibition Service, Jun-Jul 2026*

Guided prompt builder where visitors co-create AI artwork with the museum character Raimi. Ran on the exhibition floor through June and July 2026, with generation volume measured through its own backend.

- Generated **3,049** images in June 2026 and over **8,000** in July, the museum's peak season.
- Ended the unexplained freezes of the prior off-the-shelf GPT service by moving to a custom web app and serverless backend; none since.
- Designed a 10-step (+5 advanced) choice-based prompt builder that teaches prompt composition through play, for all-ages visitors.
- Auto-watermarks every image with the museum logo, storing images on Cloudflare R2 and stats in Supabase, with QR takeaway.

Tech JavaScript · OpenAI API · Vercel Serverless · Cloudflare R2 · Supabase

Doctor-Green *AI · IoT Full-Stack · Personal Project*

Smart-farm platform with live IoT environment monitoring and control in operation; its strawberry-disease diagnosis model is being rebuilt from the data up and has not been trained yet. Built solo end to end, from ESP32 sensor nodes to the deployed Next.js dashboard.

- Built the full data pipeline: ESP32 posting directly to Supabase over Wi-Fi → Next.js dashboard on 5-second polling.
- Built the training set myself: worked around AI Hub's non-Korean-IP download block with a Windows-side downloader, then extracted 39,889 lesion crops (train 32,155 / val 4,326 / test 3,408) across 5 classes (healthy plus 4 diseases) with a group-aware stratified split. Classifier training (ConvNeXt-Tiny / EfficientNetV2-S) is the next step.
- Actuator control via a desired-state pattern: the app writes target state, the device polls and applies it.

Tech Next.js 16 · TypeScript · Supabase · ESP32 · Python · Vercel · Live doctor-green-nine.vercel.app

Exhibit Auto-Recovery *Seoul Robot & AI Science Museum · Reliability & Ops*

A museum AI exhibit could only be revived by one specific person, at one specific machine. I removed the human bottleneck instead of making the restart smarter.

- Watchdog polls the exhibit every **20** seconds and restarts it within a minute of a crash or hang, with a two-step check against false positives and backoff so a deeper fault cannot become a restart loop.
- A one-tap recovery page lets any staff member restore the exhibit from their own phone, replacing the search for the one person who could.
- Designed around a failure automation cannot detect: when speech recognition dies silently, scheduled restarts and the manual button cover what the watchdog structurally cannot.
- Encrypted mesh VPN (Tailscale / WireGuard) so only registered devices reach the machine with no inbound ports exposed; handed over with a maintenance playbook.

Tech PowerShell · Windows Task Scheduler · HTTP service · Tailscale / WireGuard

ADDITIONAL PROJECTS

- **Smart-Farm Education Program:** Commissioned by the Yangpyeong Education Office through the museum; server and web-app stack built solo, finished and taught as a demo class to 5 high-school students in August 2026, with the full class scheduled for October 2026 for a 15-student cohort.
- **SEOUL RAIM Staff Platform:** Self-initiated internal app for schedules, rosters, duty status, and notices; refactored from a single-HTML prototype into Vite + React on Firebase and shipped via Capacitor, used daily by ~20 docents and part-time staff.
- **Raimi's AI Language Lab:** Five touch mini-games teaching NLP concepts (tokenization, embeddings, word order, attention, next-word prediction), bilingual (KO/EN), pure JS with no server, shipped as an offline PWA on 3 exhibition tablets.
- **I'm a Restorer!:** An interactive web game restoring Korean heritage, run on 8 tablets with ~15 participants per session.

AWARDS

- **Prize · GH Youth Build-Up Start-up Competition**

2026.03

YAKMOA, a medication-management service for digitally vulnerable users (2-person team MAKENEW; PM & Developer).

- **2nd Place • Science Museum & Community AI Hackathon** 2025.09
AI docent concept for museum–community co-prosperity (PM & Developer).
- **Encouragement Award • Chungnam Generative AI Start-up Idea Competition** 2024.10
VGG16-based Doctor-Green predecessor, hosted by the Ministry of Science and ICT (PM & Developer).
- **Excellence Award • HK StartUP&GO Audition (Hankyong National University)** 2024.10
Plant-disease diagnosis app that grew into Doctor-Green (PM & Developer).
- **Excellence Award • ICT·SW Women's Start-up Competition** 2024.12
VGG16 plant-disease diagnosis app, hosted by the Korea IT Businesswomen's Association (PM & Developer).

LEADERSHIP & ACTIVITIES

- **KB LA School (Middle School) • Middle-school Mentor** 2026.07 – Present
Dedicated mentoring, twice a week, 80 minutes a session.
- **Gwangyang AI Smart i-Kium • Instructor, Grade-3 Math & English** 2026.03 – 2026.08
Taught Grade-3 math and English once a week in Gwangyang City's AI Smart i-Kium program.
- **Geuruteogi Learning Mentoring 'Gachi-Edu' • University Mentor (6th cohort)** 2026.02 – Present
Mentoring youth in study habits and self-directed learning, twice a week, 90 minutes a session.
- **CIEE SEOULMATE • Mentor for International Students** 2025.06 – 2025.12
One-on-one Korean-language & culture mentoring; planned cross-cultural exchange programs.
- **Hankyong Start-up Club (MAKENEW) • Team Leader** 2024.07 – 2025.02
Led team MAKENEW, behind the award-winning plant-disease diagnosis app and the YAKMOA medication-care service.
- **KB LA School • High-school Mentor, Math & Chemistry (Online)** 2024.03 – 2026.02
Two years as a dedicated online mentor (Grade-11 Math I & Chemistry in 2024, Grade-10 Math in 2025), with mentees reaching top and perfect scores.

EDUCATION

Hankyong National University 2022.03 – 2027.02 (expected)
B.Eng. in Software Convergence · B.Ag. in Plant Life & Environment (double major) · completing the degree while working full-time

Rutgers University • Intensive Winter English Program 2025.01 – 2025.02
Academic English & cross-cultural communication · New Brunswick, NJ, USA

Cheongwon Girls' High School 2019 – 2022

Languages Korean (native) · English (fluent, TOEIC Speaking AL, OPIc AL)